

PhishGuard – Phishing Detection & Analysis Platform Documentation

Introduction

PhishGuard is a full-stack Phishing Detection and Analysis Platform developed to help users identify potentially malicious URLs and phishing emails through a secure and centralized digital platform. The application enables users to scan suspicious URLs, analyze email content, detect suspicious keywords and credential requests, evaluate URLs contained within emails, and monitor their previous security scans through a centralized dashboard.

The platform integrates multiple phishing detection techniques, including URL validation, HTTPS verification, suspicious keyword detection, domain analysis, IP address detection, VirusTotal analysis, and Google Safe Browsing checks. Email analysis further examines urgency indicators, credential requests, suspicious keywords, and links contained within email content.

The project demonstrates enterprise-level backend development using Java, Spring Boot, Spring Security, JWT Authentication, PostgreSQL, and RESTful APIs. The application follows a layered architecture that emphasizes scalability, maintainability, security, and modular business logic while implementing industry-standard REST API development practices.

Design Inspiration

This project is inspired by modern cybersecurity and phishing detection platforms such as:

1. VirusTotal

- URL security scanning
- Malware and threat detection
- Multi-engine analysis
- Security intelligence

2. Google Safe Browsing

- Malicious URL detection
- Phishing website detection
- Threat classification
- Safe browsing verification

3. PhishGuard Custom Implementation

- JWT Authentication
- URL Analysis
- Email Analysis

- Risk Score Calculation
- Suspicious Keyword Detection
- HTTPS Verification
- Domain Analysis
- IP Address Detection
- VirusTotal Integration
- Google Safe Browsing Integration
- Scan History
- Search and Filtering
- Security Reports
- PostgreSQL Database
- Role-Based Authentication

Roles

User

- Register account
- Login securely
- Scan suspicious URLs
- Analyze suspicious emails
- View URL scan results
- View email analysis results
- View risk scores
- View scan history
- Search previous scans
- Filter scans by type
- Filter scans by status
- View security reports

Administrator

- Login securely
- Manage users
- Monitor security activity
- View scan information
- Monitor URL scans
- Monitor email scans
- Review detection statistics
- Access system analytics

Security Administrator / Future Enhancement

- Manage detection rules
- Configure suspicious keyword lists
- Manage external security integrations

- Monitor system-wide security reports
- Review advanced threat analytics

Use Cases of the Project

Key Use Cases

User Registration

User creates an account.

Password is securely encrypted using BCrypt.

User information is stored in PostgreSQL.

User Authentication

User logs in using registered credentials.

Credentials are validated by Spring Security.

A JWT token is generated after successful authentication.

Protected APIs are accessed using:

Authorization: Bearer <TOKEN>

URL Scanning

User enters a suspicious URL.

PhishGuard performs multiple security checks:

- URL validation
- HTTPS verification
- Suspicious keyword detection
- Domain analysis
- IP address detection
- VirusTotal analysis
- Google Safe Browsing analysis

The collected results are processed by the risk scoring system.

The URL is classified as:

SAFE
SUSPICIOUS
MALICIOUS

Email Analysis

User provides:

- Sender
- Subject
- Email body

PhishGuard analyzes the email for:

- Suspicious keywords
- Urgent language
- Credential requests
- Suspicious URLs

Extracted URLs are passed through the existing URL analysis system.

The final risk score is calculated based on the detected indicators.

Scan History

Users can access their previous scans.

The history system provides:

- URL scan history
- Email scan history
- Unified scan history
- Search
- Type filtering
- Status filtering

Security Reports

The report system provides summarized security statistics including:

- Total scans
- URL scans
- Email scans
- Safe scans

- Suspicious scans
- Malicious scans
- Average risk score

Tech Stack

Backend

Java

- Core application development
- Object-oriented programming
- Business logic implementation

Spring Boot

- REST API development
- Dependency Injection
- Layered architecture
- Business logic implementation
- Application configuration

Spring Security

- Endpoint protection
- Authentication management
- Authorization handling
- Secure API access
- Role-based authorization

JWT (JSON Web Token)

- Stateless authentication
- Secure API communication
- Authorization header validation
- Token-based access control

Spring Data JPA / Hibernate

- ORM implementation
- Repository abstraction
- Entity management
- Database operations
- Relationship management

PostgreSQL

Used for persistent storage of:

- Users
- URL scans
- Email scans
- Scan results
- Authentication-related information

BCrypt Password Encoder

- Password hashing
- Secure password storage

External Security APIs

VirusTotal

PhishGuard integrates VirusTotal for URL security analysis.

VirusTotal provides additional threat intelligence that is used as one of the signals in the URL risk scoring process.

The integration analyzes:

- Malicious detections
- Suspicious detections
- Harmless results
- Analysis status

Google Safe Browsing

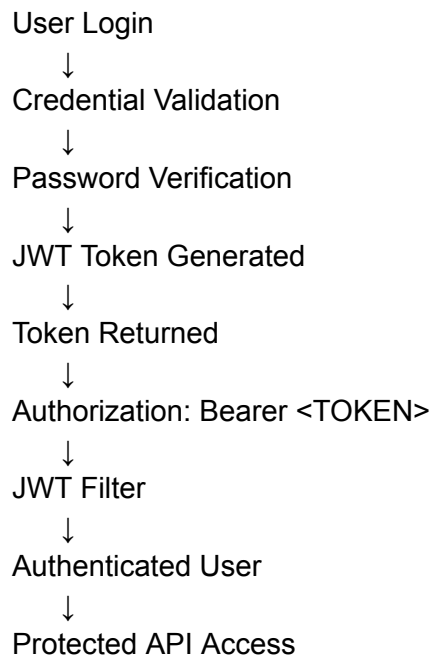
Google Safe Browsing is integrated to check URLs against Google's Safe Browsing threat detection service.

The system uses the result as an additional security signal when calculating the URL risk score.

Authentication

Authentication is implemented using JWT (JSON Web Token).

Flow



Security Layers

JwtAuthenticationFilter

The JWT authentication filter:

- Extracts the JWT token
- Validates the token
- Extracts the user's email
- Finds the authenticated user
- Verifies the user account
- Creates the Spring Security authentication context
- Protects secured endpoints

Spring Security

Responsible for:

- Endpoint protection
- Authentication
- Authorization
- Security context management
- Role-based access control

Protected APIs require authentication.

BCrypt Password Encoder

Passwords are never stored as plain text.

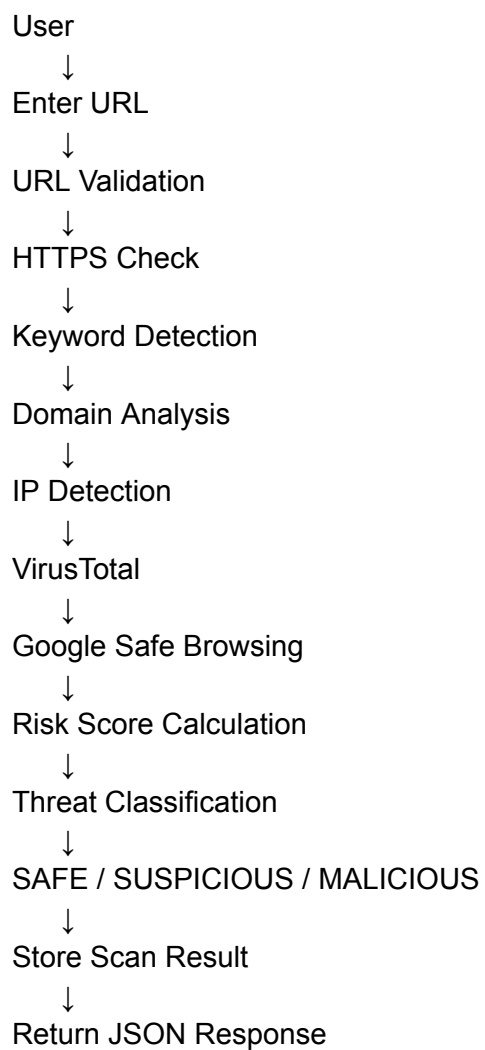
BCrypt is used for:

- Password hashing
- Secure credential storage
- Password verification

URL Detection System

The URL Scanner is one of the core modules of PhishGuard.

URL Analysis Flow



URL Detection Features

URL Validation

The system first checks whether the submitted URL is valid.

Invalid URLs are classified separately and are not passed through the external security analysis services.

HTTPS Check

PhishGuard checks whether the URL uses HTTPS.

Example:

`https://example.com`

The HTTPS result contributes to the overall risk score.

Suspicious Keyword Detection

The system checks URLs for potentially suspicious keywords.

Examples include:

login
signin
verify
verification
account
update
secure
security
password
credential
confirm
bank
wallet
payment
invoice
reset
authenticate
authorization

A suspicious keyword contributes to the URL risk score.

Domain Analysis

PhishGuard analyzes the domain extracted from the URL.

The system checks characteristics such as:

- Domain structure
- Domain length
- Suspicious domain patterns
- Excessive hyphen usage

IP Address Detection

The system detects URLs that directly use IP addresses instead of conventional domain names.

Example:

`http://192.168.1.100/login`

The system can also resolve the domain to its IP address.

VirusTotal Analysis

The submitted URL is sent to VirusTotal.

The returned analysis contributes to the overall risk assessment.

The system processes:

- Analysis Status
- Malicious Detections
- Suspicious Detections
- Harmless Results

Google Safe Browsing

The URL is checked using Google Safe Browsing.

The system evaluates whether a threat is detected.

Example result:

Threat Detected: false

Threat Type: null

Risk Score System

PhishGuard calculates a risk score using multiple security indicators.

The URL risk score considers:

- HTTPS
- Suspicious Keywords
- Suspicious Domain
- IP Address
- VirusTotal Result
- Google Safe Browsing Result

The score is limited to:

0 - 100

Risk Classification

0 - 29

SAFE

30 - 69

SUSPICIOUS

70 - 100

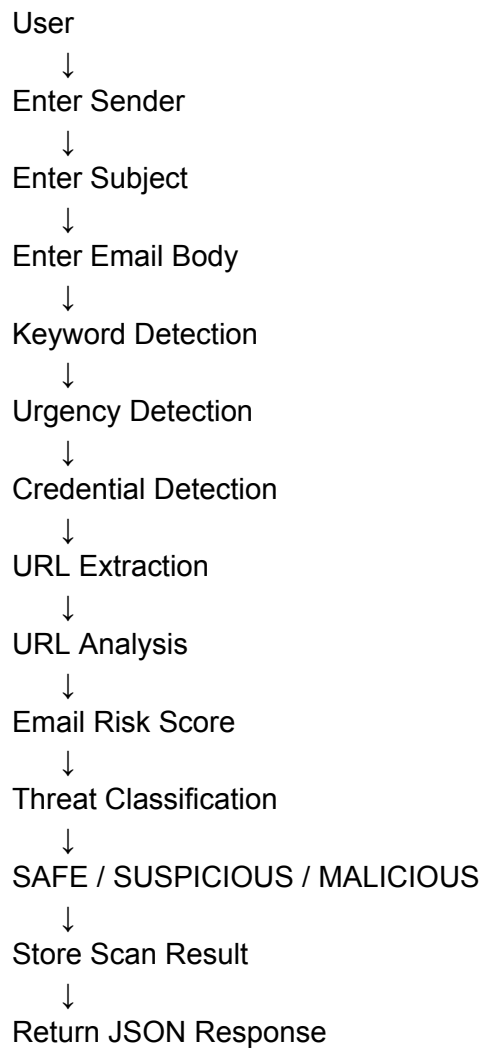
MALICIOUS

The final classification is returned to the client together with the scan results.

Email Analysis System

The Email Analyzer allows users to submit suspicious email content for analysis.

Email Analysis Flow



Email Detection Features

Suspicious Keyword Detection

The existing keyword detection system is reused to analyze the email subject and body.

This provides a common detection mechanism between URL and email analysis.

Urgency Detection

PhishGuard detects language commonly associated with urgent or threatening requests.

Examples include:

urgent
immediately

act now
action required
last warning
final warning
final notice
expires today
expires soon
within 24 hours
within 48 hours
account will be closed
account will be suspended
account will be deleted
respond immediately

Credential Detection

The system detects requests for sensitive credentials and financial information.

Examples include:

password
username
login credentials
account credentials
verify your password
enter your password
confirm your identity
security code
otp
one time password
pin
credit card
card number
cvv
bank account
banking details

URL Extraction

The email analyzer extracts URLs from the email body.

Example:

<https://example.com/login>

Extracted URLs are then analyzed using the existing URL analysis service.

This allows the platform to reuse the URL detection pipeline instead of implementing duplicate URL analysis logic.

Email Risk Score

The email risk score is calculated using:

- Suspicious Keywords
- Urgency Detection
- Credential Detection
- Highest URL Risk Score

The final score is limited to:

0 - 100

The email is then classified as:

SAFE
SUSPICIOUS
MALICIOUS

Scan History

PhishGuard provides a centralized scan history system.

Instead of creating a separate history database, the system reuses the existing:

UrlScan
EmailScan

entities.

Unified Scan History

The history API combines URL and email scan records into a single response format.

Each history record contains:

- ID
- Type
- Target
- Risk Score
- Status

- Scanned At

Example:

```
{  
  "id": 1,  
  "type": "URL",  
  "target": "https://example.com",  
  "riskScore": 0,  
  "status": "SAFE",  
  "scannedAt": "2026-09-25T12:30:00"  
}
```

For email scans, the target represents the sender.

Search

Users can search their scan history.

Example:

GET /api/history?search=example

For email scans, searching can match:

- Sender
- Subject

For URL scans, searching can match:

- URL

Filtering

The history system supports filtering by scan type.

GET /api/history?type=URL

or:

GET /api/history?type=EMAIL

Users can also filter by status:

GET /api/history?status=SAFE

GET /api/history?status=SUSPICIOUS

GET /api/history?status=MALICIOUS

Multiple filters can be combined.

Example:

GET /api/history?search=security&type=EMAIL&status=SUSPICIOUS

Security Reports

PhishGuard provides a report endpoint for summarized scan statistics.

GET /api/history/reports

The report provides:

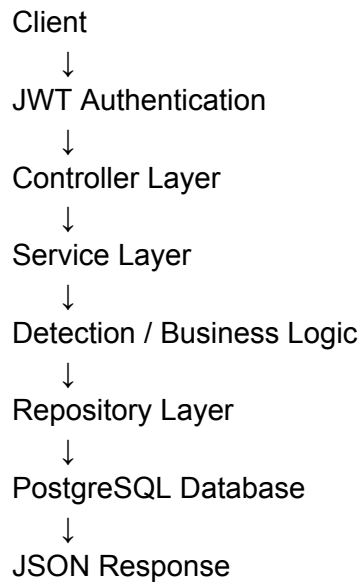
- Total Scans
- URL Scans
- Email Scans
- Safe Scans
- Suspicious Scans
- Malicious Scans
- Average Risk Score

Example structure:

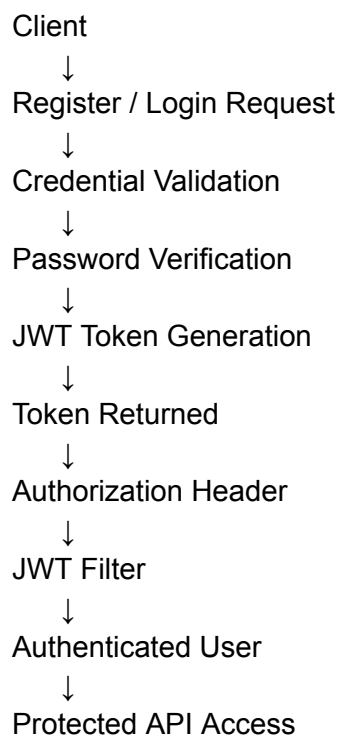
```
{
  "totalScans": 20,
  "urlScans": 12,
  "emailScans": 8,
  "safe": 10,
  "suspicious": 6,
  "malicious": 4,
  "averageRiskScore": 38.75
}
```

Flowcharts

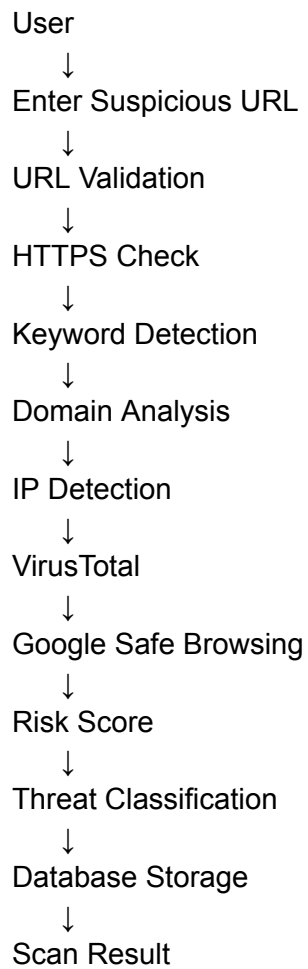
Backend Flow



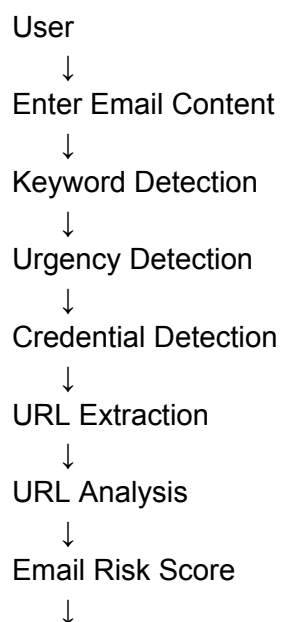
Authentication Flow

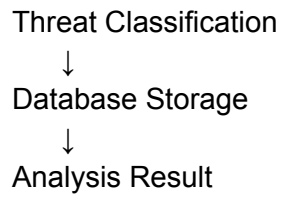


URL Scanning Flow

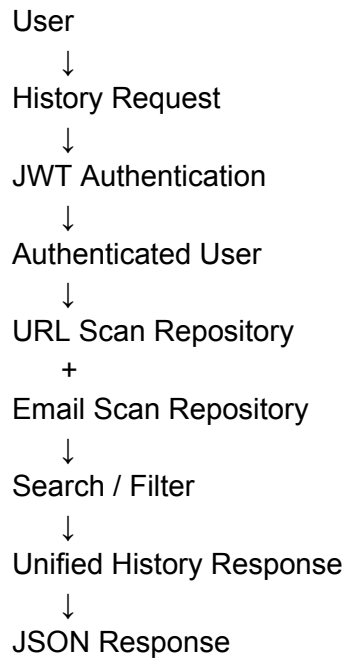


Email Analysis Flow

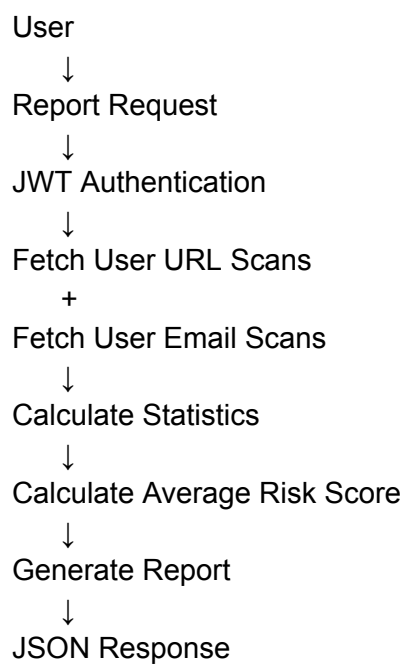




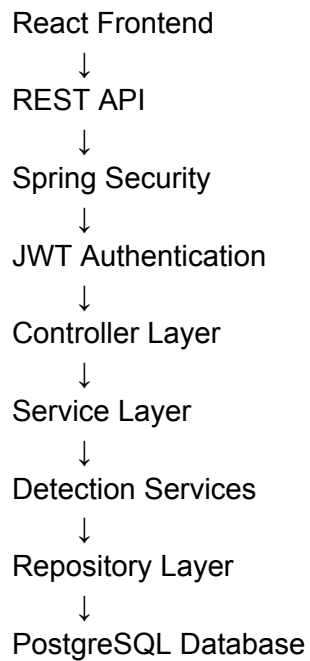
Scan History Flow



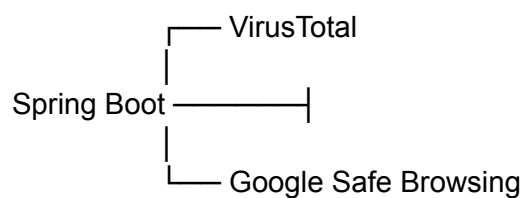
Report Flow



System Architecture

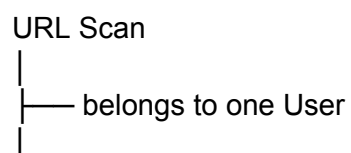
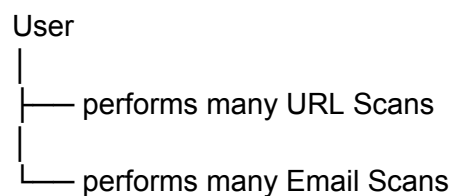


External security integrations:



Database Design

ER Concept



- contains URL analysis results

Email Scan

— belongs to one User

More specifically:

User

```
graph LR
    Root[ ] --- UrlScan[< UrlScan]
    Root --- EmailScan[< EmailScan]
```

User

The User entity stores authenticated user information.

It contains information such as:

- ID
- Name
- Email
- Password
- Role
- Created At
- Active Status

Passwords are stored using BCrypt hashing.

URL Scan

The URL scan entity stores the results of URL analysis.

The scan record includes information such as:

- ID
- URL
- Status
- Scanned At
- HTTPS
- Risk Score
- Suspicious Keyword
- Suspicious Domain

- IP Address
- Resolved IP
- User

Email Scan

The email scan entity stores the results of email analysis.

The record includes:

- ID
- Sender
- Subject
- Risk Score
- Status
- Suspicious Keyword
- Urgency Detected
- Credential Detected
- Scanned At
- User

Frontend Integration Flow

React Dashboard



REST API Calls



Spring Boot Backend



JWT Authentication



URL / Email Analysis



PostgreSQL Database



JSON Response



Dashboard Update

The frontend communicates with the backend through REST APIs and sends the JWT token through the Authorization header when accessing protected endpoints.

API Integration

Authentication

POST /api/auth/register

POST /api/auth/login

URL Scanner

POST /api/url/scan

URL history:

GET /api/url/scans

Individual URL scan:

GET /api/url/scans/{id}

Email Analyzer

POST /api/email/scan

Email history:

GET /api/email/scans

Individual email scan:

GET /api/email/scans/{id}

Unified Scan History

GET /api/history

Search:

GET /api/history?search=example

Filter by type:

GET /api/history?type=URL

Filter by status:

GET /api/history?status=SUSPICIOUS

Combined filtering:

GET /api/history?search=security&type=EMAIL&status=SUSPICIOUS

Reports:

GET /api/history/reports

Industry Value of the Project

PhishGuard demonstrates real-world cybersecurity and enterprise software development concepts.

Industry Applications

- Cybersecurity Platforms
- Phishing Detection Systems
- Security Operations Tools
- Email Security Platforms
- Threat Intelligence Systems
- Enterprise Security Portals
- Security Awareness Platforms

The architecture can be extended to support additional security detection engines, threat intelligence services, administrative analytics, and advanced security monitoring.

Key Concepts Implemented

- JWT Authentication
- Spring Security
- Role-Based Authorization
- RESTful API Development
- Layered Architecture
- DTO-Based Design
- CRUD Operations
- URL Analysis
- Email Analysis
- Risk Score Calculation
- Suspicious Keyword Detection
- HTTPS Detection
- Domain Analysis
- IP Detection
- VirusTotal Integration
- Google Safe Browsing Integration
- Scan History

- Search and Filtering
- Security Reports
- PostgreSQL Database Design
- Global Exception Handling
- Request Validation
- Dependency Injection
- Spring Data JPA
- Hibernate
- BCrypt Password Hashing

Technologies Used

Backend

- Java 21
- Spring Boot
- Spring Security
- JWT
- Spring Data JPA
- Hibernate
- PostgreSQL
- Maven
- BCrypt Password Encoder
- REST APIs

External APIs

- VirusTotal API
- Google Safe Browsing API

Frontend

- React
- Tailwind CSS
- Axios / Fetch API
- Vite

Integration Overview

React Frontend



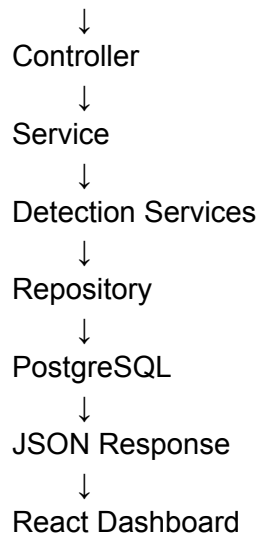
REST API



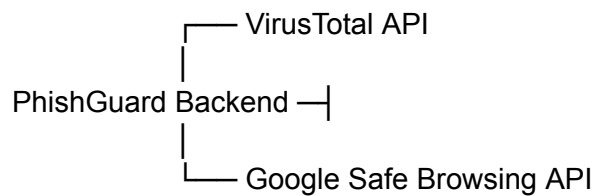
Spring Security



JWT Authentication



External security services:



Error Handling and Validation

PhishGuard implements centralized exception handling for consistent API responses.

The application handles situations such as:

- Invalid Request
- Validation Errors
- Invalid Scan Requests
- Scan Not Found
- Authentication Errors
- Authorization Errors

Request validation is implemented using Jakarta Validation.

For example, required fields such as email sender, subject, and body are validated before processing.

Data Security

Security is implemented at multiple levels.

Password Hashing



BCrypt



JWT Authentication



Spring Security



Authenticated User



Protected API



User-Specific Database Queries

User-specific scan history is associated with the authenticated User, ensuring that history queries operate against the current user's records.

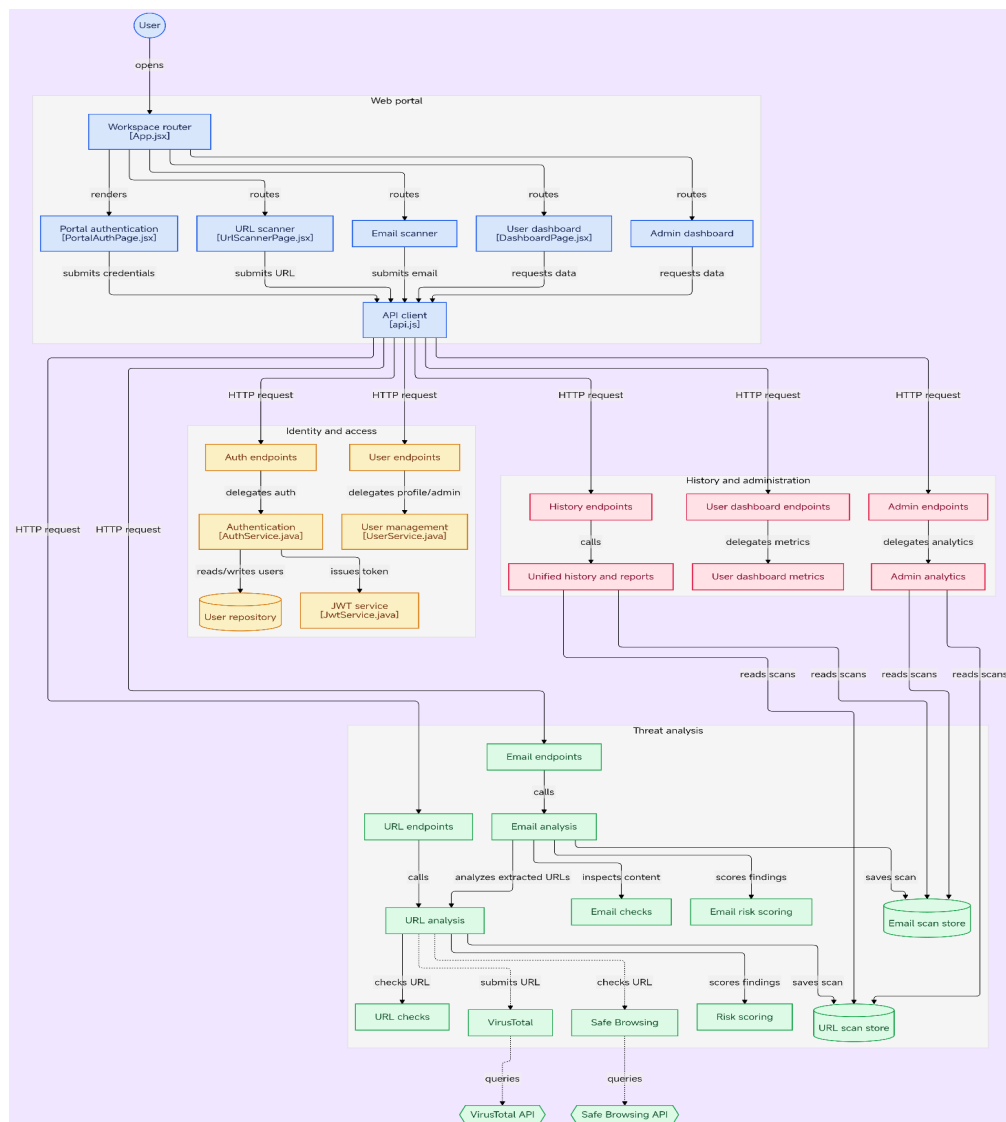
Conclusion

PhishGuard successfully demonstrates the design and implementation of a secure, modular, and scalable Phishing Detection and Analysis Platform.

Key Achievements

- Secure JWT Authentication
- Spring Security Integration
- Role-Based Authorization
- URL Phishing Detection
- Email Phishing Analysis
- Suspicious Keyword Detection
- Urgency Detection
- Credential Request Detection
- HTTPS Verification
- Domain Analysis
- IP Address Detection
- VirusTotal Integration
- Google Safe Browsing Integration
- Risk Score Calculation
- SAFE / SUSPICIOUS / MALICIOUS Classification
- URL Scan History
- Email Scan History
- Unified Scan History

- Search and Filtering
- Security Reports
- PostgreSQL Relational Database
- Global Exception Handling
- Request Validation
- RESTful API Development
- Layered Software Architecture
- DTO-Based Design



PhishGuard combines multiple phishing detection techniques into a centralized security platform. The URL scanner evaluates suspicious links using local detection rules and external threat intelligence services, while the Email Analyzer identifies suspicious language, credential requests, and potentially dangerous URLs contained within email messages.

The platform also provides unified scan history, search, filtering, and security reports, allowing users to monitor their previous security analyses from a centralized system.

PhishGuard showcases industry-standard backend development practices using Java, Spring Boot, Spring Security, JWT Authentication, PostgreSQL, Spring Data JPA, REST APIs, DTO-based architecture, external security APIs, and role-based access control, making it a practical cybersecurity-focused full-stack project and a strong demonstration of enterprise backend development